

# PRANOY P JYOTHIRAJ

Mechanical Engineering • Quality Management & TQM • Process Improvement

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## EXECUTIVE SUMMARY

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Final-year Mechanical Engineering student graduating July 2026 with hands-on experience in quality systems, process documentation, and cross-functional coordination at an automotive component manufacturer. Work involves identifying process requirements from OEM customer drawings, building SOPs and Work Instructions, conducting dimensional inspections, and driving corrective actions through Quality Alert documentation. Also built a web application that identified and eliminated a manual process inefficiency in the QA team. Comfortable working across production, quality, and engineering teams to ensure process compliance. Looking to apply this hands-on engineering and quality background in a Graduate Engineer Trainee role within the automotive manufacturing sector.

## EDUCATION

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**B.E. Mechanical Engineering – Sri Venkateswara College of Engineering** 2022 – 2026

CGPA: 8.06 / 10.0 | Relevant: Engineering Drawing, Manufacturing Processes, Quality Engineering, Machine Design, Thermodynamics

**Class XII – TI Matriculation Higher Secondary School** 2021 – 2022

Percentage: 83%

## PROFESSIONAL EXPERIENCE

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**Quality Assurance Intern** March 2026 – Present

MRP AutoRub Pvt. Ltd. – Automotive Rubber Hose Manufacturer Kakkalur, Tiruvallur

- Identified process requirements from OEM customer drawings (Daimler, Ashok Leyland, Tata Motors, Mahindra, Modine, Schwing Stetter) and developed SOPs and inspection criteria, averaging 8–10 SOPs a day with consistent manager sign-off.
- Applied GD&T principles and tolerance tables to calculate min/max dimensional ranges for ID, OD, wall thickness, bend radii, and profile lengths across 250+ active runner parts.
- Prepared Quality Alert documents for process deviations and developed corrective action plans in coordination with the production team.
- Drafted bilingual (English/Tamil) Work Instructions for printing, pad marking, and packing operations, used directly on the shop floor by operators.
- Identified and eliminated a manual data entry inefficiency by building a web application that automated the QA team's daily inspection report process.
- Maintained traceability records and supported cross-functional coordination between quality, production, and management teams.

**In-Plant Intern** June 2024 – July 2024

Blasto Metal Spray Processors Pvt. Ltd. Chennai, TN

- Observed machining, fabrication, and thermal spray processes for wind turbine components; understood how manufacturing constraints and surface treatments affect component performance.

**Design & Development Intern** January 2024

National Small Industries Corporation (NSIC) Chennai, TN

- Training exposure to PTC Creo and SolidWorks for solid modelling and part design.

## PROJECTS

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**IC Engine Performance & Emission Study – B20 Biodiesel with Al<sub>2</sub>O<sub>3</sub>-doped CeO<sub>2</sub> Nanoparticles**

- Synthesised Al<sub>2</sub>O<sub>3</sub>-doped CeO<sub>2</sub> nanoparticles and prepared a stable nano-dispersed B20 blend (20% Jatropha biodiesel + 80% diesel) using ultrasonication to prevent particle settling.
- Set up and ran load-based experimental trials on a 4-stroke single-cylinder direct injection CI engine across five load conditions (0%, 25%, 50%, 75%, 100%).
- Measured BTE, BSFC, CO, HC, NO<sub>x</sub>, and smoke opacity using a flue gas analyser and AVL smoke meter across all test conditions.
- Compared results across nanoparticle concentrations and identified the optimal blend for fuel efficiency and emission reduction against a neat diesel baseline.

## SKILLS

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**Quality & TQM:** Quality Management Systems, SOP Development, Work Instructions, Quality Alerts, Inspection Planning, Root Cause Analysis, Corrective Action, Traceability Documentation, Process Improvement

**Engineering:** GD&T, Tolerance Analysis, Dimensional Metrology, Engineering Drawing Interpretation, Cross-functional Coordination

**Software:** MS Office 365 (Excel – Advanced), PTC Creo (Basic), SolidWorks (Basic), Python (Basic), Web Application Development

**Soft Skills:** Structured Problem Solving, Attention to Detail, Team Collaboration, Communication across functions

## LEADERSHIP ROLES

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### **General Secretary – Forum for Economic Studies by Engineers (FORESE), SVCE**

*AY 2025–2026*

- Runs club operations, drives decision-making within the core panel, and leads student engagement programs across the department.

### **Treasurer – Forum for Economic Studies by Engineers (FORESE), SVCE**

*AY 2024–2025*

- Managed club finances, tracked expenses, and set up basic financial controls for club operations.